Modeling How Macroeconomic Shocks Affect Regional Employment: Analyzing the Brazilian Formal Labor Market using the Global VAR Approach

Journal:	*Journal of Regional Science*
Manuscript ID	Draft
Manuscript Type:	Original Manuscript
Keywords:	GVAR, cointegration, regional dynamics, labor markets

Modeling How Macroeconomic Shocks Affect Regional Employment: Analyzing the Brazilian Formal Labor Market using the Global VAR Approach

April 25, 2019

Abstract

Assessing linkages across regions and how macroeconomic shocks spread out across regions is not an easy task. We address the curse of dimensionality using a global vector autoregressive methodology. Focusing on the Brazilian labor market, identified and quantified how a shock in an aggregate economic activity spreads out regionally and throughout time. Another novelty is the use of information collected by the Brazilian Bureau of Geography and Statistics to measure how regions are linked by analyzing infrastructure linkages municipalities. We provide evidence that macroeconomic shocks cause stronger effects on more developed regions where formal labor market is better developed.

JEL codes: R11, E27, C23.

Key words: GVAR, cointegration, regional dynamics and labor market.

1 Introduction

Macroeconomic shocks affect sectors and regions with different intensities and delays. Population and economic activities tend to cluster in specific areas due economies of scale and agglomeration. The population tends to locate near the sea coast compared with interior continental areas. Sector activities tend to be concentrated in some areas or cities. Existence of a good economic infrastructure such as roads, airports, ports, and telecommunications facilities links regions economically and provides channels that can spread out economic shocks among regions. Macroeconomic models tend to disregard regional and spatial economic activity issues due to either their aggregate nature bias or complexity of theme. However for a specific region, it is important to understand how economic activity variance can be decomposed into macroeconomic and idiosyncratic factors.

In the past years, time series data at disaggregated and regional levels are becoming available. At the same time, efforts have been made in econometric literature to model high-dimensional problems. Two challenges must be faced to address interdependence among regions, and linkages from regions to macroeconomic scenario. The first is the curse of dimensionality. The second challenge involves modeling the linkage between regions.

Chudik and Pesaran [2011] suggested two approaches to handle the curse of dimensionality: (i) shrinkage of the parameter space and (ii) shrinkage of the data. The authors dealt with the curse of dimensionality by shrinking the parameter space as the number of endogenous variables tends to infinity. In this configuration, the infinite-dimensional vector autoregressive model could be well approximated by a set of finite-dimensional small-scale models. Moreover, this model can be estimated separately in a consistent manner, which is the global vector autoregressive (GVAR) model proposed by Pesaran et al. [2004].

Therefore, this study proposes a novel approach for dealing with the curse of dimensionality when modeling the labor market: use of the GVAR methodology to model the labor market at a regional level, accounting for interdependence among regions as well as among global macroeconomic variables. The GVAR methodology eliminates the curse of dimensionality and provides a parsimonious model for the labor market, and since the all variables are treated as endogenous, the GVAR model allows the global variables to be affected by the regional variables through a feedback channel.

We use information collected by the Brazilian Institute of Geography and Statistics (IBGE)[1] that maps infrastructure and economic linkages across all municipalities in Brazil to construct the weights used in GVAR to model interdependence across regions. We do not restrict our analysis to neighborhood effects on economic activity by constructing a measure of economic connections.

Focusing on the Brazilian labor market, the aim of study is twofold. First, we want to model the link between an aggregate macroeconomic variable and regional labor dynamics as well as accounts for interdependence among Brazilian regions. To achieve that, a GVAR methodology proposed by Pesaran et al. [2004] is used. Second, we want to map how regional formal labor employment reacts to Brazilian national economic activity. We employ monthly records of admissions and discharges in the Brazilian formal regional labor market. The period of the study is only 2004 to 2016 due to data constraints. The model analysis led to the conclusion that the Brazilian Northeast and South regions are the most elastic, whereas the North and Central regions are less sensitive to macroeconomic shocks. The Southeast region - the most developed in the country - is at the middle level.

This study is divided into sections as follows: section 2 is a brief literature review of previous works related to the purpose of this study; section 3 presents the methodology; section 4 defines the models used; section 5 describes the data with a brief description of sources and characteristics; section 6 reports the results; and section 7 contains final considerations.

[1]Initials come from "Instituto Brasileiro de Geografia e Estatísticas".

2 Literature Review

This section presents a brief literature review on the literature that deals with spatial time series models and labor market.

Firstly, focusing on the Brazilian labor market, Oliveira and Carneiro [2001] analyzed the fluctuations in the employment level of several Brazilian states compared to national employment. The purpose of their study was to verify whether it is possible to establish a long-term relationship between state employment and nationwide employment. To achieve this the authors applied a cointegration analysis, a methodology proposed by Engle and Granger [1987], as well as the model of unrestricted error correction as proposed by Pesaran et al. [1996].

The authors used a database that included a comprehensive coverage of the labor market in the Brazilian states, using data from the Annual Report on Social Information (RAIS)[2] from 1985 through 1996. The results obtained from an Vector Error Correction model were consistent with those obtained with Engle and Granger method (Engle and Granger [1987]). The results lead to the conclusion that employment fluctuations in most states shares a common trend with national employment level.

The hypothesis that the formal labor market presents a different dynamics in Brazilian metropolitan and non-metropolitan areas was tested by Kretzmann and da Cunha [2009]. The goal of their study was to analyze the fluctuations in the labor market of metropolitan and non-metropolitan Brazilian areas. They used employment and unemployment data from the Ministry of Labor and Employment through the General Records of Employed and Unemployed Persons - CAGED[3] - using monthly data from 1996 through 2007. Kretzmann and da Cunha [2009] used cointegration analysis through the traditional method proposed by Engle-Granger, as well as the method proposed by Pesaran et al. [1996]. The two methods produced the sim-

[2]RAIS from the Portuguese Relação Anual de Informações Sociais.

[3]Initials comes from "*Cadastro Geral de Empregados e Desempregados*" in Portuguese)

ilar results, suggesting that the metropolitan and non-metropolitan areas are not co-integrated.

None of these paper deals with the curse of dimensionality. The authors opt to with pairs of series. The evidence that states and national share a common trend highlight a possible channel linking macro and regional dynamics. Evidence that metropolitan and non metropolitan areas do not share a common trend may be explained by the fact that neighborhood effects are not that relevant but when corrected by infrastructure connections they may be. This hypothesis is investigated in this work.

Secondly, Pesaran et al. [2004] introduced the Global Vector Autoregressive (GVAR) methodology by means of a study that modeled 11 different regions using data from 1979 through 1999. This methodology was improved by Dees et al. [2007] with the collaboration of the European Central Bank.

One of the first studies that applied GVAR to tackle interdependence across regions is Schanne [2015]. The author analyzed the regional labor market in Germany with a focus on forecasting regional labor markets. His study had two goals: the first one is to estimate a the GVAR model, as proposed by Pesaran et al. [2004]. Second, to use the developed GVAR model to construct a forecast for German employment accounting for spatial effects.

The GVAR model allows both the analysis of strong cross-section dependence and weak cross-section dependency. Consequently, Schanne [2015] concludes that the data indicate semi-strong cross-section dependence and has consistent results since Germany is a polycentric economy, in contrast with the United Kingdom or France (places where there are clearly dominant regions). The final model used in his study is a basic GVAR specification, which is a model in first differences without imposing cointegration relations, and did not include a dominant region.

In his forecasting experiment, Schanne [2015] estimated the basic GVAR model and GVAR models with leading indicators. The author used these models as a predictive forecasting tool for three and twelve months ahead – forecasting the level of unemployment in the regional divisions according to the federal employment agency of Germany. For comparison purposes, each indicator was estimated by a VAR in differences and a VAR in level. The accuracy of the prediction was evaluated by comparing the absolute mean of prediction errors (percentage) and by the Diebold-Mariano test (Diebold and Mariano [1995]). Finally, Schanne [2015] concluded that there is significant improvement in the accuracy of predictions when considering the interdependence between regions, emphasizing that the degree of improvement depends on the variables and horizon.

3 Methodology

This section presents and describes the GVAR methodology, which is the main methodology used in this study. It is assumed that the reader is familiar with the VAR methodology and VECM as well as their terminologies.

Chudik and Pesaran [2011] considered the curse of dimensionality in the case of large linear dynamic systems. To deal with the curse, they proposed to shrink the parameter space in the limit as the number of endogenous variables tends to infinity. They showed that by shrinking the parameter space, the infinite-dimensional VAR could be well approximated by a set of finite-dimensional small-scale models that can be consistently estimated separately, which is in the spirit of the GVAR models proposed in Pesaran et al. [2004]. Therefore, Chudik and Pesaran [2011] concluded that the GVAR approach can be used as an approximation to an nfinite dimension vector autoregressive model (IVAR) featuring all macroeconomic variables. This is true for stationary models as well as for systems with variables integrated of order one.

The GVAR model is a compact model that combines individual VECMs in which domestic variables are related to individual-specific foreign variables in a consistent manner.

3.1 Global vector autoregressive model

The GVAR methodology was originally proposed by Pesaran et al. [2004] as a practical approach to construct a coherent global model. In the traditional VAR methodology, an unrestricted VAR(p) model with k endogenous variables covering N countries has a large number of unknown parameters of the order $(k.N)^2p + kN$ in the mean plus $\frac{(k.N+1)kN}{2}$ in the variance, thus requiring a parsimonious solution. Therefore, the authors proposed the use of a model wherein, first, the error correction models specific to each region are estimated, with the domestic macroeconomic variables related to the corresponding foreign variables, and second the models from each region are combined to generate forecasts for all variables. In short, the GVAR model combines the VECM models of each region, so that domestic variables are related to regionally specific foreign variables in a consistent manner. Therefore, the model can consider the interdependencies between economic regions in the analysis. Subsequently Dees et al. [2007] improved the GVAR model with the cooperation of the European Central Bank. This version of the GVAR model included 26 countries (covering 90% of world production), with the Euro area being considered as one sole economy with estimates from 1979 to 2003. The version of GVAR model used in this study is based on that of Dees et al. [2007].

The GVAR model can be summarized as a two-step procedure. In the first step, specific models of each region are estimated that are conditioned to the external influences of other regions. That is, the models are represented as VAR models, which have domestic variables and foreign variables, the latter being treated as weakly exogenous.[4] In the second step, the VAR models of each region are grouped into one single GVAR model. The GVAR model, as well as any VAR model, can be used for analyzing shocks in the economy and making predictions.

3.1.1 Specific models for each region

For the specific models used in the first stage of the GVAR methodology, consider a

[4] See Engle et al. [1983] for a definition.

panel with N elements, each element showing k_i variables observed during periods $t = 1, 2, \cdots, T$. Within this panel structure the following variables are defined: $x_{i,t}$ as a vector $k_i \times 1$ of domestic variables for element i in period t; x^*_{it} as a vector $k^* \times 1$ of foreign variables; and $u_{i,t}$ as an error vector. Thus, following the notation of Chudik and Pesaran [2016], the specific models of each element consist of a set of domestic and foreign variables. Specifically, for any i element:

$$x_{i,t} = a_{i0} + a_{i1}t + \sum_{j=1}^{p_i} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x^*_{i,t} + \sum_{l=1}^{q_i} \Lambda_{il} x^*_{i,t-l} + u_{i,t}, \tag{1}$$

where Φ_{ij} is an array of lag coefficients for lag j associated with domestic variables, and Λ_{il} is an array of lag coefficients for lag l associated with foreign variables.

The GVAR model treats foreign variables as weights averages of the domestic variables of each corresponding element, and the weights are specific to each unit. Therefore, the foreign variables of each element i can be computed as weights of the domestic variables of other units in the panel. W_{ij} is a weighting matrix, where $j = 1, 2, \cdots, N$ is defined as a set of weights with the following characteristics: $W_{ii} = 0$ and $\sum_{j=1}^{N} W_{ij} = 1$. Thus, the foreign variables can be defined as

$$x^*_{i,t} = \sum_{j=0}^{N} W_{ij} x_{jt}. \tag{2}$$

The weights are predetermined and used to capture the importance of region j for the economy of region i.

From the domestic and foreign variables estimated from the weighting previously mentioned, VECMs are estimated separately for each region, considering that in the VECMs, the foreign variables $x^*_{i,t}$ are treated as weakly exogenous. The use of the VECMs allows cointegration between domestic variables and between domestic and foreign variables.

3.1.2 Global model

Although estimation is performed individually, element by element, the GVAR model is solved by considering all variables as endogenous to the system. Thus, from the specific model of each element, $Z_{i,t}$ is defined as

$$Z_{i,t} = \begin{bmatrix} x_{i,t} \\ x^*_{i,t} \end{bmatrix}, \tag{3}$$

Using (2) in (3) it is possible to obtain

$$A_{i0} Z_{i,t} = a_{i0} + a_{i1} t + \sum_{l=1}^{p} A_{il} Z_{i,t-l} + \varepsilon_{i,t}, \tag{4}$$

where $p = \max_i\{p, q_i\}$, $A_{i0} = [\ I_{k_i} \quad -\Lambda_{i0}\ ]$, and $A_{il} = [\ \Phi_{il} \quad \Lambda_{il}\ ]$ for $l = 1, 2, \cdots, p$. Moreover we define $\Phi_{il} = 0$ for $l > p_i$ and $\Lambda_{il} = 0$ for $l > q_i$.

Let $x_t = [x'_{0,t}, x'_{1,t}, \cdots, x'_{N,t},]'$ be a vector $k \times 1$ of all the variables in the panel, where $k = \sum_{i=1}^{N} k_i$. W_i can be defined as a matrix of dimension $(k_i + k_i^*) \times 1$ of known constants defined in terms of the individual weights W_{ij}. Therefore, variables of the specific models are defined as a function of x_t.

$$Z_{i,t} = W_i x_t \tag{5}$$

Using this identity, the individual models can be written as

$$A_{i0} W_i x_t = a_{i0} + a_{i1} t + \sum_{l=1}^{p} A_{il} W_i x_{t-l} + \varepsilon_{i,t}, \tag{6}$$

It is possible to combine the new individual models into one single model, and this can be done by "stacking up" each model in a large-scale VAR.

$$G_0 x_t = a_0 + a_1 t + \sum_{l=1}^{p} G_l x_{t-l} + \varepsilon_t, \tag{7}$$

where

$$G_0 = \begin{bmatrix} A_{00} W_0 \\ A_{10} W_1 \\ \vdots \\ A_{N0} W_N \end{bmatrix}, \quad G_1 = \begin{bmatrix} A_{0l} W_0 \\ A_{1l} W_1 \\ \vdots \\ A_{Nl} W_N \end{bmatrix}, \quad a_0 = \begin{bmatrix} a_{00} \\ a_{10} \\ \vdots \\ a_{N0} \end{bmatrix}, \quad a_1 = \begin{bmatrix} a_{01} \\ a_{11} \\ \vdots \\ a_{N1} \end{bmatrix}, \quad \varepsilon_t = \begin{bmatrix} \varepsilon_{0,t} \\ \varepsilon_{1,t} \\ \vdots \\ \varepsilon_{N,t} \end{bmatrix},$$

Since the G_0 matrix is a non-singular matrix, which depends only on the estimated parameters and the link matrix W_i, it is possible to rewrite equation (7) as

$$x_t = b_0 + b_1 t + \sum_{i=1}^{p} F_i x_{t-i} + \epsilon_t, \tag{8}$$

where

$$F_i = G_0^{-1} G_i,$$
$$b_0 = G_0^{-1} a_0,$$
$$b_1 = G_0^{-1} a_1,$$
$$\epsilon_t = G_0^{-1} \varepsilon_t,$$

Hence, the global model, which consists of a set of individual models from each region, can be solved recursively. With that, it is possible to consistently determine the coefficients of the model.

3.1.3 Dominant unit model

Global VAR is based on the hypothesis of weak exogeneity ([Engle et al., 1983]). If this hypothesis fails for one or more regions, these regions must be modeled jointly. In spatial analysis, it is possible that some important regions affect others in a specific form. This information must be included in the model. Therefore, vector ω_t of common variables and its lagged values augment each specific model. So, each element of the model can be stated as

$$x_{i,t} = a_{i0} + a_{i1} t + \sum_{j=1}^{p_i} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x^*_{i,t} + \sum_{l=1}^{q_i} \Lambda_{il} x^*_{i,t-l} + D_{i0}\omega_t + \sum_{l=1}^{s_i} \Psi_{il} \omega_{t-l} + u_{i,t}, \quad (9)$$

The marginal model for the dominant variables can be estimated with the feedback effects from x_t, or not. The feedback from the variables in the GVAR model are captured via cross-section averages, and this is done by augmenting the dominant unit model with lags of $x^*_{\omega t}$.

$$\omega_t = \sum_{l=1}^{p_\omega} \Phi_{\omega l} \omega_{t-l} + \sum_{l=1}^{q_\omega} \Lambda_{\omega l} x^*_{i,t-l} + \eta_{\omega t} \quad (10)$$

Different lag orders for the dominant variables (p_ω) and cross-section averages (q_ω) could be considered. It is worthy to point out that contemporaneous values of foreign variables do not feature in the equation.

3.2 Regional weights

Finally, the number of connections between regions were obtained from IBGE [2008]. This work aims to determine economic linkage among municipalities in Brazil. The study analyzed not only regular transport links, such as those that go to the urban centers, but also the main destinations of the residents to obtain products and services, such as general goods, education, airport travel, and health services. The acquisition of agricultural inputs and destination of the agricultural products were also analyzed in the study. The main result of the research is a Brazilian urban network map that connects all 5,564 Brazilian municipalities. This map contains the hierarchy of the urban network and the regions of influence of each urban center. A chart indicating the connections between all Brazilian cities is constructed by merging the links provided in the urban network framework, which are the main links between municipalities, with the links regarding airport connections, shopping destinations, education, health services, and recreation activities.

Since every city belongs to a mesoregion, the links between mesoregions are also determined with the following characteristics: if the two cities are within the same region then the link between them is an internal link; if the two cities are in different mesoregions, then the link between them is a connection between the two mesoregions.

As mentioned earlier, the weight matrix is a function of the number of connections between regions; however at this point, it is worth mentioning that some connections are more important than others. Moreover, since the measurement of importance for each connection between each region is unfeasible, an equal weight to all connections is assumed, resulting in a linear function, which is the average.

The weight matrix to region i can then be determined by the number of connections to other regions divided by its total number of connections. Therefore, the weight vector for each region is determined as

$$W_i = \frac{\begin{bmatrix} C_{i1} & \cdots & C_{iN} \end{bmatrix}}{\sum_{j=1}^{N} C_{ij}}, \tag{11}$$

where C_{ij} is the number of connections that originated at region i and have region j as destination, keeping in mind that connections within the same region $(i = j)$, also called internal links, have a zero weight ($C_{ii} = 0$).

One most important step in GVAR methodology is the construction of the weight matrix. In Pesaran's initial work, the weight matrix is defined by country-specific trade weights. In fact, the weight matrix W_i is designed to capture the importance of all other regions for region i (Di Mauro and Pesaran, 2013). So, to determine the weight matrix it is necessary to determine the importance of each region in respect to every other region.

Focusing on the labor market, to determine the importance of a region to another is to determine the importance of one labor market to another labor market. Each region has its own labor market with firms and workers, and since firms cannot change markets, it is the workers that freely change markets. In other words, a worker can look for a job in other regions, but a firm must attract workers from other regions to its own market. Consequently, for a worker to commute to another region, there must be a connection between these regions.

The connections between regions go beyond neighborhood effects in an integrated economy. There are many other channels such as physical infrastructure that allow workers to commute (by car, by train, by plane, etc.) to another region. Therefore, regions with higher connections are more important than regions with fewer connections to model a specific unit.

Consequently, the importance of region j to region i will be a function of the connections between them, and the weight matrix will be a based on the number of connections between the regions.

4 Database Description and Model Specification

The labor market interacts with the macroeconomic environment through many channels. One of these channels is the level of economic activity. Industrial production is used as a proxy to economic activity. It is highly correlated to the gross domestic product (GDP) but available at a higher frequency level. The first is available quarterly whereas the latter is available monthly.

Finally, an attempt to model the Brazilian labor market at the municipality level was made but then discarded. Brazil had a total of 5,564 municipalities in 2005 according to IBGE. This government institute grouped municipalities into 137 mesoregions and 27 states. Many municipalities were merged or split, and thus it is difficult to construct a long time series. Therefore, we opt to work at the mesoregional level.

4.1 Database

Table 8 contains a detailed description of the data sources. The following section provides a brief description of the data used in this study as well as their sources and frequency.

<Insert Table 8 about here>

Data for the Brazilian industrial production were obtained from IBGE. The employment and unemployment records, which indicate the workers admitted to and discharged from the labor market, were obtained from CAGED.[5] The data were obtained at the municipality level and grouped by mesoregions. The frequency of the data is monthly starting from January 2004 to December 2016.

The mesoregions' definitions, as well as information on the total population, microregions, mesoregions, states, and their designation codes, were obtained from IBGE as reported in 2015.

4.2 Descriptive statistics

Table 1 shows descriptive statistics for the industrial production as well as the admitted and disconnected series for mesoregions that contain the cities of São Paulo, Rio de Janeiro, and Brasilia - three of the most important cities in Brazil. The descriptive statistics for all the regions can be found in the appendices.

<Insert Table 1 about here>

All models were estimated using MatLab software R2016b and GVAR Toolbox 2.0.

[5]Initials come from "Cadastro Geral de Empregados e Desempregados".

The following MatLab add-ons were installed during the estimation of the models: Mapping Toolbox version 4.4, Econometrics Toolbox version 3.5, Optimization Toolbox version 7.5, and Statistics and Machine Learning Toolbox version 11.0.

4.3 Global vector autoregressive model specification

The objective of this study is to use the GVAR methodology, as described in the previous section, to assess the Brazilian labor market model. Each Brazilian mesoregion will be a GVAR unit.

Since the labor market model consists of workers admitted to and discharged from the labor market, the specific model for each mesoregion will contain information of those admitted to and discharged from the labor market in period t for that region. Therefore, for each mesoregion i a vector x_{it} can be defined as

$$x_{i,t} = \begin{bmatrix} Adm_{i,t} \\ Dis_{i,t} \end{bmatrix}, \tag{12}$$

where $Adm_{i,t}$ is the number of workers admitted (employment flow) to the labor market in period t in region i, and $Dis_{i,t}$ is the number of workers disconnected (unemployment flow) from the labor market in period t in region i.

Each region experiences influences from other regions. Using the GVAR methodology, the foreign variables are defined as a vector $x^*_{i,t}$ as follows:

$$Adm^*_{i,t} = \sum_{j=1}^{N} W_{ij} Adm_{j,t}, \tag{13}$$

$$Dis^*_{i,t} = \sum_{j=1}^{N} W_{ij} Dis_{j,t}, \tag{14}$$

$$x^*_{i,t} = \begin{bmatrix} Adm^*_{i,t} \\ Dis^*_{i,t} \end{bmatrix}, \tag{15}$$

where W_{ij} is a link matrix that captures how much region j influences region i; $Adm^*_{i,t}$ is the influence on region i of the number of workers admitted to the labor market in other regions in period t; and $Dis^*_{i,t}$ is the influence in region i of the number of workers disconnected from the labor market in other regions in period t; The specific models are defined with unrestricted intercepts and no trend coefficients. It is worth mentioning that even though a deterministic trend is not included in the model, the levels of the domestic variables may be trended due to the drift coefficients.

$$x_{i,t} = a_{i0} + \sum_{j=1}^{k_{1,i}} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x^*_{i,t} + \sum_{l=1}^{k_{2,i}} \Lambda_{il} x^*_{i,t-l} + u_{i,t} \tag{16}$$

where $k_{1,i}$ and $k_{2,i}$ are the lag order for the domestic variables and foreign variables respectively. The model is defined with a maximum two lag order for domestic and foreign variables, where the lag selection for each region will be done by the GVAR Toolbox using the Akaike information criteria.

4.4 Macroeconomic dynamics

In the GVAR model, macroeconomic variables are treated as common variables and modeled in a dominant unit. Therefore, the dominant unit of the GVAR model will contain the log level of the Brazilian industrial production defined by the vector ω_t, with a feedback variable from the regions through the aggregate values of employment and unemployment. So, the dominant unit will be a univariate model defined in first differences with a trend included in the levels, and it can be expressed as

$$\omega_t = \left[\log(IndProd_t) \right], \tag{17}$$

$$\Delta\omega_t = \Phi_{\omega 0} + \sum_{l=1}^{k_1} \Phi_{\omega l} \Delta\omega_{t-l} + \sum_{l=1}^{k_2} \Lambda_{\omega l} \Delta x^*_{i,t-l} + \eta_{\omega t}, \tag{18}$$

where Δx^* is the feedback variable containing the aggregate values of employment and unemployment from all regions; k_1 and k_2 are the lag orders for the global variable (industrial production) and feedback variables, respectively. The maximum lag order is defined as three for both the global variable and feedback variables, where the lag selection for each variable will be set by the GVAR Toolbox using Akaike information criteria.

5 Results

Modeling a system such as the Brazilian labor market at the mesoregion level is naturally subject to considerable uncertainties. There are many choices to be made for each individual mesoregion model, for example, about the variables to be included in the specific models, lag order of the domestic variables, lag order of the foreign variables, number of cointegrating relations, and whether to impose long-run and short-run restrictions on the parameters.

As explained earlier, the GVAR model comprises several specific models, solved in a two-step process. The model is defined using the data for admissions and discharges for every mesoregion as well as the weight matrix defined previously. All regions were set to be affected by the foreign variable and dominant unit.

The final model was estimated using the GVAR automatic lag selection with the Akaike information criterion and a maximum lag order of two for domestic as well as foreign variables. A total of 137 models were specified. Lags for domestic variables were set to two in 98 models and to one in the other 39 models. The lags for foreign variables were set to two in 91 models and to one in the other 46 models.

Existence of cointegration among variables was investigated using an automated procedure of GVAR Toolbox 2.0. All individual models had the deterministic components treated as an unrestricted intercept and no trend in the error correction models. Out of the 137 estimated models, nine were specified with one cointegration relationship, and 128 models were specified with two cointegration relationships.

The GVAR model determines a model for each region considering foreign effects as well as effects from the variables in the dominant unit model. With no overidentifying restrictions imposed, the final model proved to be stable with 815 eigenvalues lying inside the unit circle, 10 of which were lying on the unit circle.

5.1 Forecast exercise

Using the GVAR methodology, it is possible to estimate the dynamic response of unemployment and employment in each region, considering not only the industrial production but also the interdependence of each region. With the use of conditional forecasts, it is possible to evaluate the behavior of the labor market in response to a change in the global economy, which in our case is represented by the industrial production. Therefore, a series of conditional forecasts for different growth scenarios of industrial production were made to assess the impact of the industrial production on the labor market. The series of conditional forecasts of the labor market consist of a base scenario and several other growth scenarios for the Brazilian industrial production.

The baseline scenario for industrial production growth will have 0% growth coefficient over the last year in the series, in this case, 2016. The other growth scenarios on the set will have 2% growth coefficient, 4% growth coefficient, 6% growth coefficient, and 8% growth coefficient over the last year.

The industrial production growth is determined by applying the annual growth coefficient in each month compared with that of the last year. In other words, a 2% growth coefficient means a 2% grow in January 2016 compared to January 2017 and so on. All the scenarios were analyzed over an eight-year forecast.

5.1.1 Forecast, net employment, and accumulated employment

"Net employment gain" is defined as the difference between admissions minus discharges. In addition "total net employment mesoregion gain" is defined as the sum of the net employment gain for every municipality of the mesoregion. Consequently, as the forecast exercise started in January 2017, the cumulative employment gain is the sum of all the previous "total net employment" until that year. The total cumulative employment is an aggregate of all mesoregions and can be used to evaluate how the Brazilian labor market will behave in different macroeconomic scenarios in terms of changes in stocks of employment. (Figure 1)

<Insert Figure 1 about here>

From 2015 through 2016, the total net employment gain is -3 million because Brazilian economy was facing a deep recession. Taking December 2016 as reference, the baseline forecasts reveal that a 0% annual growth for industrial production would imply a great loss of formal employment. If industrial production starts to recover at the rate of 2% annual growth, then the net gain will become positive only about September of 2021. At the end of the forecast horizon period, a cumulative gain of 2.5 million net jobs will be created. A 4% annual growth in the industrial production would cause significant improvement to the cumulative net employment gain with the first positive net gain of the series occurring in September 2017, and the cumulative gain of employment will be 6.5 million by the end of 2024. For 6% and 8% annual growth, cumulative gain will become positive in June 2017, and it will have a value of 10.5 million and 14.5 million, respectively, by the end of the forecast period. It is worth mentioning that 4% annual growth in the industrial production would compensate for the Brazilian cumulative net employment deficit from 2015 to 2016 only in July 2022.

5.2 Industrial production elasticity by region

The net employment defined in the last section can be interpreted as the variation on the total employment level. To normalize the value we divide each variable by total population of each mesoregion. This gives an idea about the dynamics of each region. It would be better to have an estimate of labour force, but this is not available. Therefore, changes in the employment rate used in this study are a proxy of employment rate creation.

Since the GVAR methodology allows for a forecast for each region, it is possible to map changes of the employment rate for every region, where the cumulative changes are the sum of all previous variations. Therefore, it is possible to evaluate how a change in the industrial production affects the labor market in each region; in other words, it is possible to establish the employment elasticity with respect to the industrial production. (Figure 2)

<Insert Figure 2 about here>

Data on each mesoregion are grouped into five Brazilian regions: North, Northeast, Southeast, South and Centralwest. Taking the IBGE's estimated population for 2016 as the total population, and 0% growth as a base scenario, it is possible to obtain the industrial production elasticity of employment for each major region over an eight-year forecast period. Considering a 2% annual growth in the industrial production, the South region is the most sensitive with 0.18 elasticity; the second most sensitive region is the Northeast with 0.15 elasticity. The Southeast region has 0.10 elasticity, while the least sensitive major regions are the North and Centralwest, each with 0.06 elasticity.

This indicates that the Northeast and South regions have higher elasticity, being strongly affected by changes in industrial production. The Southeast region, which is the most developed, has intermediate elasticity. This is probably due to the high development level of the region, making its marginal gain not as high as in the Northeast and South regions.

Although previous analysis was conducted in macroregions, the GVAR methodology allows us to conduct the same analysis at a more in-depth level. Since the GVAR equations are solved for every region, we can evaluate elasticity for not only the state but also for every mesoregion.

Figure 3 shows the estimated employment rate change for the Southeast region, the most developed in the country. Using 0% growth as a base scenario, elasticity for the states of Minas Gerais, Espírito Santo, Rio de Janeiro, and São Paulo are estimated. Again, considering a 2% annual growth in industrial production, Rio de Janeiro is the most sensitive state with 0.19 elasticity, followed by Espírito Santo with a 0.09 elasticity. Minas Gerais and São Paulo are estimated to have elasticity of 0.07 and 0.08, respectively.

<Insert Figure 3 about here>

As mentioned earlier, with the GVAR model, it is possible to evaluate every region in a more in-depth analysis. Since listing all 137 regions' estimated elasticity would be impractical, the region of São Paulo metropolitan area (one of the most important in the country) is compared with four other randomly selected regions: Goiás Central region, Borborema region, Jequitinhonha region, and Porto Alegre metropolitan area. Goiás Central region is the most rich and densely populated mesoregion of the Goiás state, with 3.1 million inhabitants. The mesoregion of Borborema is dry land, with approximately 310,000 inhabitants. The Jequitinhonha Valley region is known for its low social indicators. It has 731,000 inhabitants. Porto Alegre metropolitan area is an important region in the South and its economy is strongly influenced by the industrial sector 5 million inhabitants.

Figure 4 shows the estimated elasticity for the regions. As previously mentioned, 0% growth is used as a base scenario for the elasticity. With the scenario of 2% annual growth in industrial production, Porto Alegre is the most sensitive mesoregion with 0.19 elasticity, followed by São Paulo with 0.16 elasticity. The Goiás central area has a moderate elasticity with a value of 0.07. Lastly, the Borborema and Jequitinhonha mesoregions proved to be very inelastic, with only 0.02 elasticity.

<Insert Figure 4 about here>

The most populated mesoregions in Brazil are São Paulo metropolitan area, Rio de Janeiro metropolitan area, Belo Horizonte metropolitan area, Porto Alegre metropolitan area, and Salvador metropolitan area. According to IBGE in 2017, Rio de Janeiro metropolitan area has approximately 12.9 million inhabitants, and Belo Horizonte metropolitan area has approximately 6.7 million inhabitants, and Salvador metropolitan area has around 4.7 million inhabitants.

Figure 5 shows the estimated elasticity for the most populated mesoregions in Brazil. As before, 0% growth is used as a base scenario for the elasticity. With the scenario of 2% annual growth in industrial production, the metropolitan areas of Rio de Janeiro and Porto Alegre are the most sensitive mesoregions with 0.19 elasticity. Belo Horizonte and São Paulo have elasticity of 0.17 and 0.16, respectively. Lastly, the metropolitan area of Salvador has 0.11 elasticity.

<Insert Figure 5 about here>

5.3 Regional dynamics

One advantage of the GVAR methodology is the ability to detect dynamic interactions between the regions in the model. In other words, the GVAR model can provide a forecast with the dynamics of each region. Consequently, the final analysis of our study evaluates the interactions among the regions in different scenarios.

We discuss the dynamics of employment in four scenarios for up to seven years ahead. The initial period is December of 2016. Scenarios to be compared are 0% growth (baseline), 2% growth, and 4% growth in industrial production, and the forecast horizons will be one year, three years, five years, and seven years ahead.

It is expected that the most populated areas will have higher number of admissions and discharges, or higher net employment, than less populated areas. Therefore, to account for the total population in the region, the comparison between the regions will be made using the previously defined employment rate.

By using the GVAR methodology and comparing the accumulated employment rate, it is possible to determine not only how the employment rate in each region reacts to a variation in industrial production, but also dynamics of the regions' interactions.

The South, Southeast, and Northeast regions proved to be very dynamic, displaying constant interaction within its regions, while the North and Central regions appeared to be little influenced by neighboring regions.

As shown in Figure 6, the 0% growth scenario shows almost no improvement in the individual regions. In December 2017, this scenario has 112 mesoregions, which show negative net employment. In December 2023, only 28 regions develop a positive balance of employment, and 84 mesoregions still show negative net employment (the 84 regions are divided as follows: North: 15; Northeast: 33; Southeast: 16; South: 14; and Centralwest: 6). The 2% growth scenario shows some improvement over the 0% growth, with only 41 mesoregions having negative net employment in December 2023 (North: 14; Northeast: 18; Southeast: 6; South: 1; and Centralwest:

2). Finally, the 4% growth scenario shows significant improvement; only 13 among all 137 mesoregions have negative net employment by December 2023 (North: 6; Northeast: 3; Southeast: 2; South: 1; and Centralwest: 1).

<Insert Figure 6 about here>

6 Limitations and possible extensions

This work can be expanded in many forms. First, a more comprehensive macroeconomic model can be formulated and different types of macroeconomic shocks can be identified and their impact simulated. Second, it is possible to construct a "tailor-made" GVAR. Each individual model can be estimated using recently developed techniques for model selection such as autometrics (Hendry and Doornik, 2014). This route was explored by Ericsson and Reisman [2012] but not in a regional model. It is also possible to disaggregate data by not only regions but also sectors such as agriculture, industry and services among others.

Another possible path of research consists of evaluating predictive performance at different aggregate levels and horizons against simple benchmarks such as random walk or first order autoregressive model.

7 Conclusion

The initial purpose of this study was twofold: first, creating a model of the Brazilian labor market at a regional level using the GVAR methodology, as proposed by Pesaran et al. [2004] and second, using the developed model to estimate the regional elasticity of employment with respect to the economic activity of the country while identifying the dynamics of the regions as well as inferring on the effects of the global variables on the labor market.

The increasing integration of the Brazilian territory has generated great interdependence among its regions. This interdependence plays a key role in the modeling and analysis of the markets. Therefore, macroeconomic analyses should consider the existence of a correlation not only between local and global variables but also with variables from other regions through this interdependence channel.

In this scenario, the GVAR methodology, developed by Pesaran et al. [2004], plays a leading role in the modeling of the labor market, since it accounts for the interdependence and connections between economies and regions. Therefore, if the assumptions of weak exogeneity are maintained, it is possible to use the GVAR to construct a simple and practical model for the labor market.

We document the existence of a cointegration relationship between regions using our indicator of economic linkage that goes beyond neighborhood effects. Using GVAR model we are able to consolidate all interdependence between regions in a comprehensive manner. The model can be used to simulate the evolution of employment under different macroeconomic scenarios and unveil and quantify linkages across regions.

Thus, using the previously estimated model, several forecast scenarios for the labor market are analyzed, leading to the assessment of a situation in which industrial production remains at the current level. In other words, for the 0% annual growth coefficient, the labor market will keep a negative employment balance, meaning that the economy will remain in recession at least for the next eight years. On the other hand, 2% annual growth of the industrial production shows significant improvement in the accumulated net employment leading to positive accumulated balance of 2.5 million employment positions for an eight year forecast. Higher growth scenarios (4%, 6%, and 8% annual coefficients) also show higher accumulated employment. Therefore, an increase in industrial production affects the labor market in a positive correlation, as expected according to the macroeconomic theory.

Focusing the main regions in Brazil – North, Northeast, Southeast, South, and Centralwest – and using the forecast scenarios, it was possible to estimate the elasticity of employment in relation to industrial production for each major region. The most sensitive region is the South followed by the Northeast; they have elasticity of 0.18 and 0.16, respectively. The Southeast, the country's most developed region, has intermediate elasticity, whereas the North and Central regions have the lowest elasticity.

Finally, results from the model suggest a rich dynamic relationship among mesoregions. The South, Southeast, and Northeast are quite dynamic regions and present higher level of interaction, whereas in the North and Centralwest regions, these links do not appear to be strong.

Finally, our study suggests that the GVAR methodology can give good insights about the dynamics of formal labor employment and deal with the curse of dimensionality in a manageable form.

References

Alexander Chudik and M Hashem Pesaran. Infinite-dimensional vars and factor models. *Journal of Econometrics*, 163(1):4–22, 2011.

Alexander Chudik and M Hashem Pesaran. Theory and practice of gvar modelling. *Journal of Economic Surveys*, 30(1):165–197, 2016.

Stephane Dees, Filippo di Mauro, M Hashem Pesaran, and L Vanessa Smith. Exploring the international linkages of the euro area: a global var analysis. *Journal of applied econometrics*, 22(1):1–38, 2007.

Filippo Di Mauro and M Hashem Pesaran. *The GVAR handbook: structure and applications of a macro model of the global economy for policy analysis.* OUP Oxford, 2013.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. URL http://www.jstor.org/stable/1392185.

Robert F Engle and Clive WJ Granger. Co-integration and error correction: representation, estimation, and testing. *Econometrica: journal of the Econometric Society*, pages 251–276, 1987.

Robert F Engle, David F Hendry, and Jean-Francois Richard. Exogeneity. *Econometrica: Journal of the Econometric Society*, pages 277–304, 1983.

Neil R Ericsson and Erica L Reisman. Evaluating a global vector autoregression for forecasting. *International Advances in Economic Research*, 18(3):247–258, 2012.

David F Hendry and Jurgen A Doornik. *Empirical model discovery and theory evaluation: automatic selection methods in econometrics*. MIT Press, 2014.

IBGE. Regioes de influencia das cidades. *IBGE*, 2008.

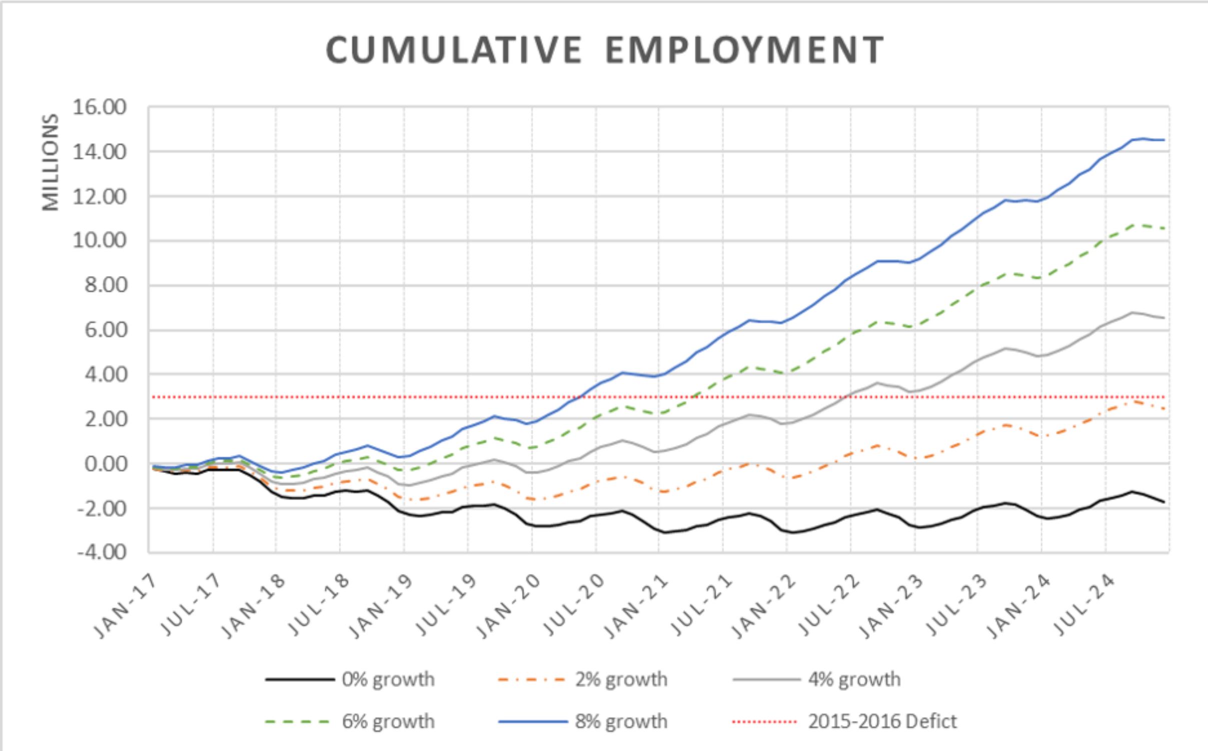
Camila Kraide Kretzmann and Marina Silva da Cunha. Flutuações no mercado de trabalho brasileiro: regiões metropolitanas e não-metropolitanas. *Revista EconomiA*, 2:401–419, 2009.

Carlos Wagner de Albuquerque Oliveira and Francisco Galrão Carneiro. Flutuações de longo prazo do emprego no brasil: uma análise alternativa de co-integração. *Revista Brasileira de Economia*, 55(4):493–512, 2001.

M Hashem Pesaran, Yongcheol Shin, Richard J Smith, et al. Testing for the'existence of a long-run relationship'. Technical report, Faculty of Economics, University of Cambridge, 1996.

M Hashem Pesaran, Til Schuermann, and Scott M Weiner. Modeling regional interdependencies using a global error-correcting macroeconometric model. *Journal of Business & Economic Statistics*, 22(2):129–162, 2004.

Norbert Schanne. A global vector autoregression (gvar) model for regional labour markets and its forecasting performance with leading indicators in germany. Technical report, IAB Discussion Paper, 2015.



Source: elaborated by the author

Note: 2015-2016 deficit: The net employment deficit accumulated from 2015 through 2016

Figure 1: Brazilian accumulated employment forecast for different growth scenarios

8 Tables and Figures:

Data	Source	Frequency	Period
Brazilian industrial production	IBGE (1)	Monthly	2004-01 to 2016-12
Employment and unemployment	CAGED (2)	Monthly	2004-01 to 2016-12
Mesoregions and municipalities	IBGE (1)	N/A	as reported in 2015
Connection between regions	IBGE (1)	N/A	as of 2007

Source: elaborated by the author

(1) Brazilian Institute of Geography and Statistics

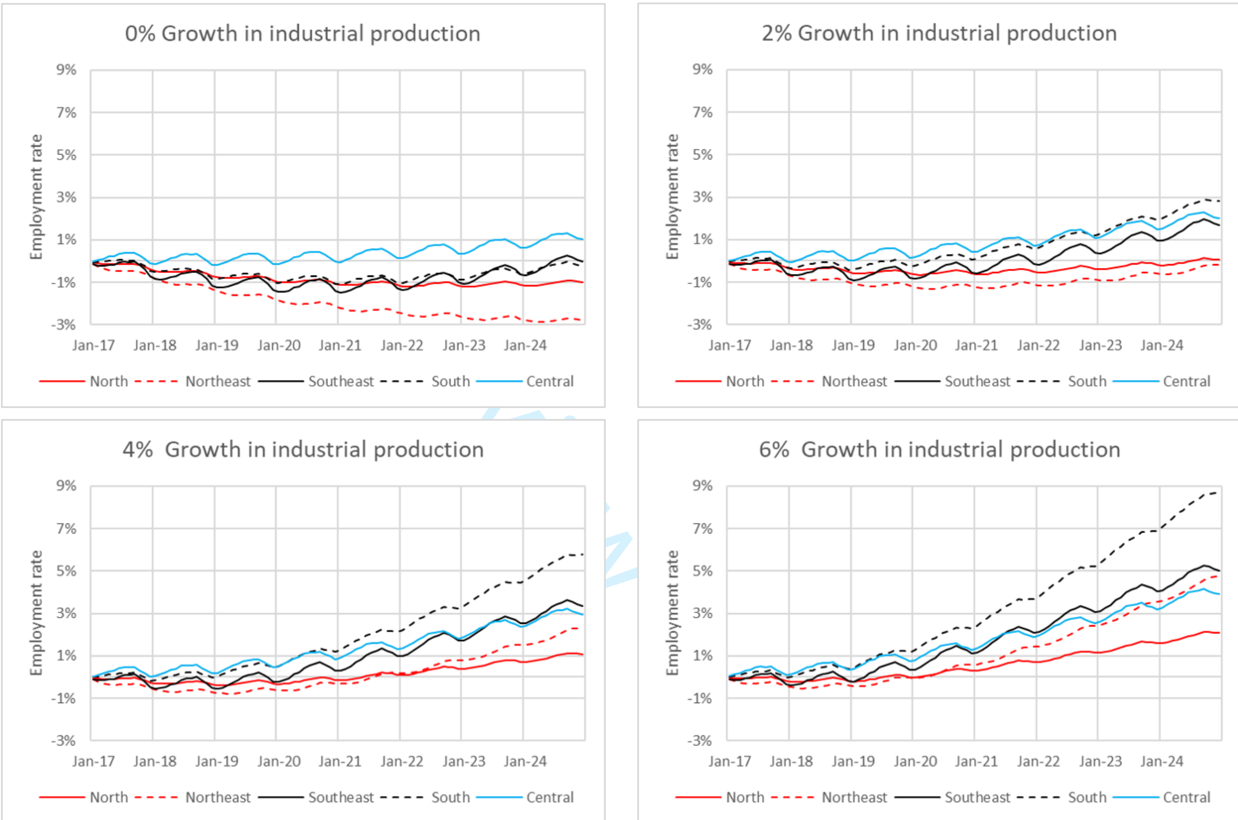
(2) Ministry of Labor and Employment through the General Records of Employed and Unemployed Persons

Table 1: Descriptive statistics of the variables

Region	Series Name	Mean	Median	Max	Min	Std. Dev.	Skewness	Kurtosis	Jarque-Bera
Global	Ind. Prod.	4.5503	4.5502	4.7238	4.3041	0.0963	-0.3792	2.59	4.7326
SP - Metropol Sao Paulo	Admitted	226073	223610.5	329823	115254	55443.5	-0.0856	1.7569	9.8751
SP - Metropol Sao Paulo	Discharged	212768.2	217592	298117	107413	56594.5	-0.307	1.8034	11.4459
RJ - Metropol do RJ	Admitted	95295.5	94490	140206	56900	22461.6	0.1119	1.7417	10.2581
RJ - Metropol do RJ	Discharged	90694.9	92642.5	141759	51785	23537.8	-0.0474	1.7966	9.1128
GO - Distrito Federal	Admitted	22942.6	22685.5	36420	12557	5643.1	0.0728	1.738	10.1272
GO - Distrito Federal	Discharged	22068.6	22648.5	33552	10988	5931.7	-0.1269	1.7852	9.6577

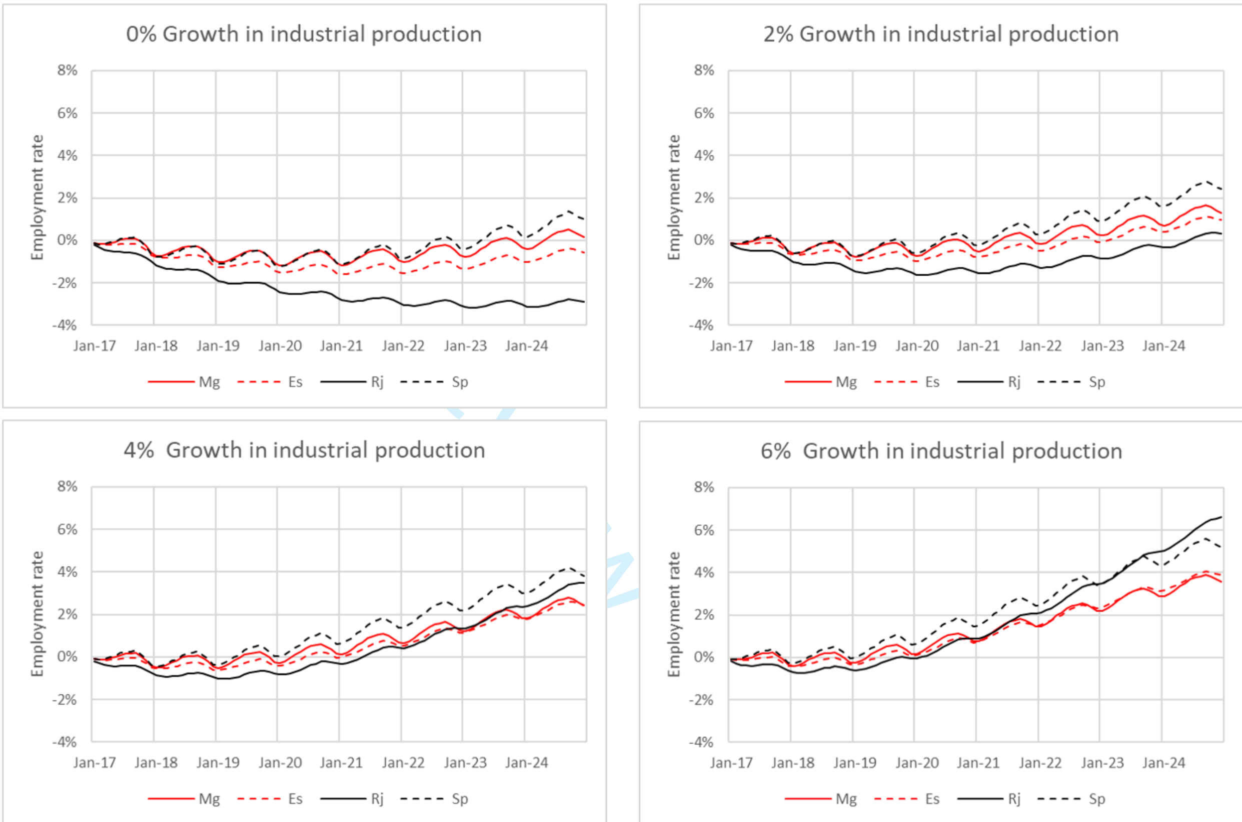
Source: elaborated by the author

Note: The data for the industrial production are presented at log level. SP denotes São Paulo State,RJ denotes Rio de Janeiro and GO denotes Goiás.



Source: elaborated by the authors

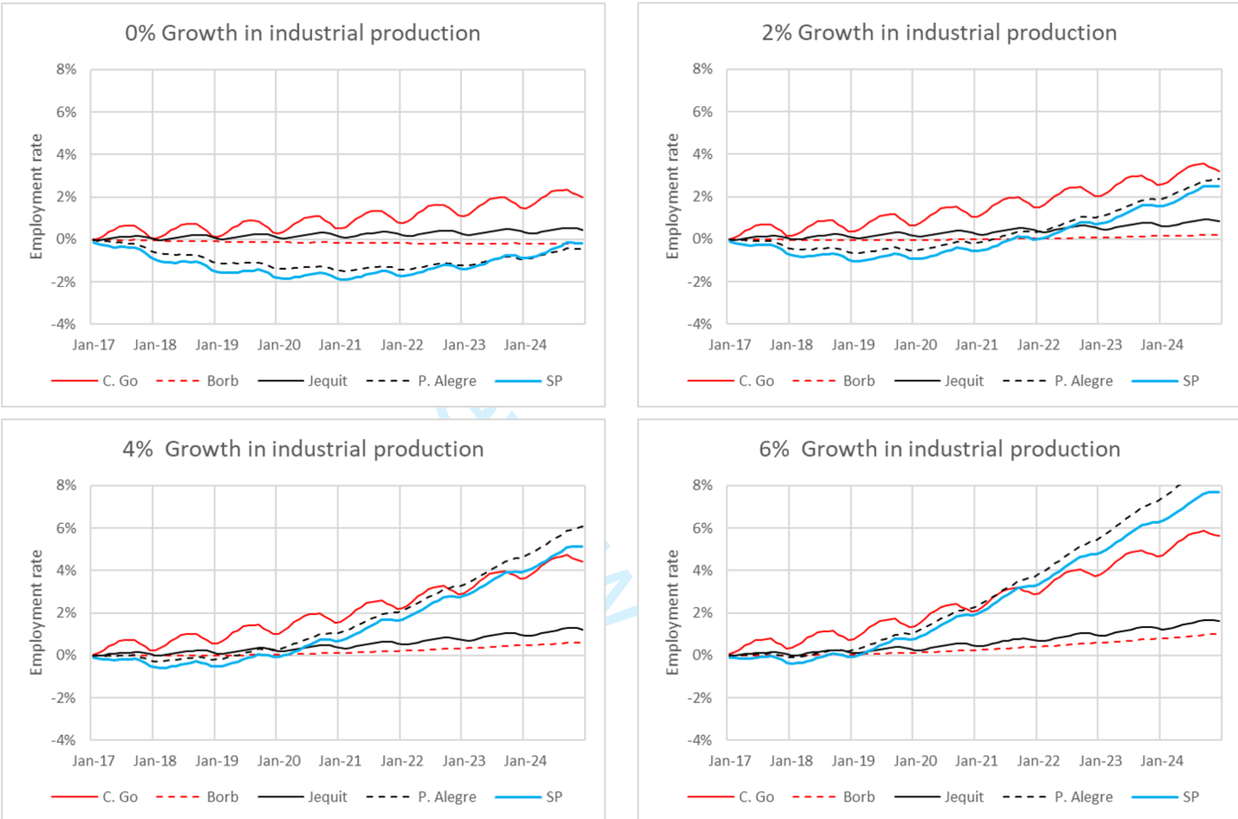
Figure 2: Accumulated employment rate variation for various industrial production growth scenarios



Source: elaborated by the author

Mg: Minas Gerais; Es: Espírito Santo; Rj: Rio de Janeiro; Sp: São Paulo.

Figure 3: Accumulated employment rate variation for various industrial production growth scenarios on state level.

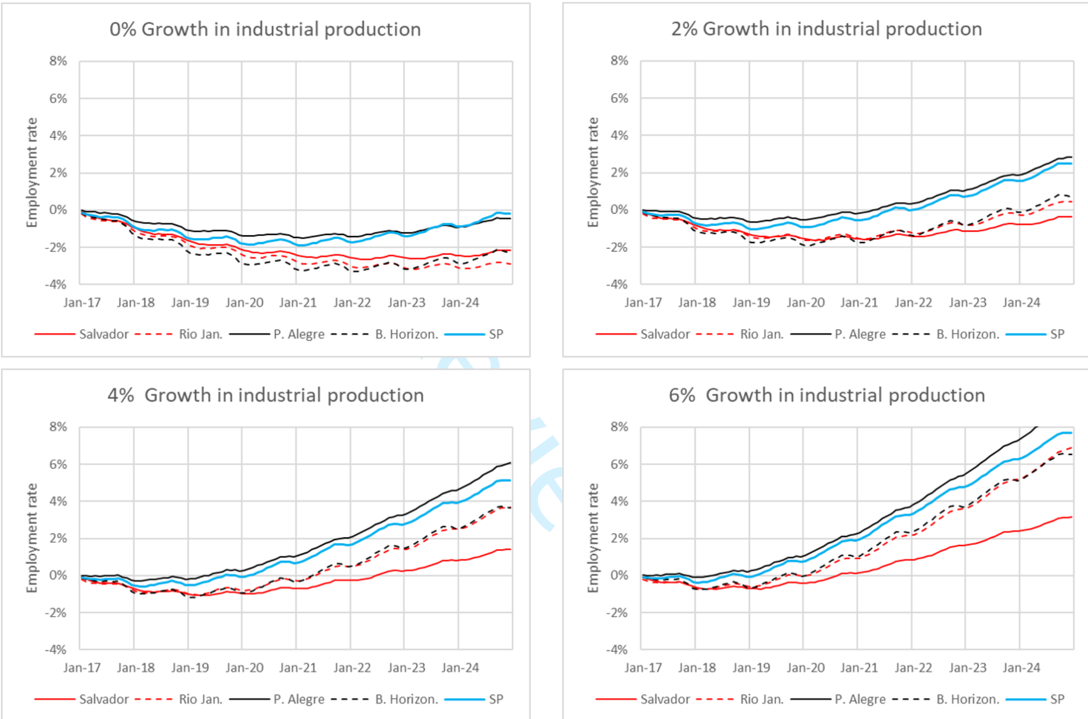


Source: elaborated by the author

C.Go: Goiás central area; Borb: Borborema; Jequit: Jequitinhonha;

P.Alegre: Porto Alegre metropolitan area; SP: São Paulo metropolitan area

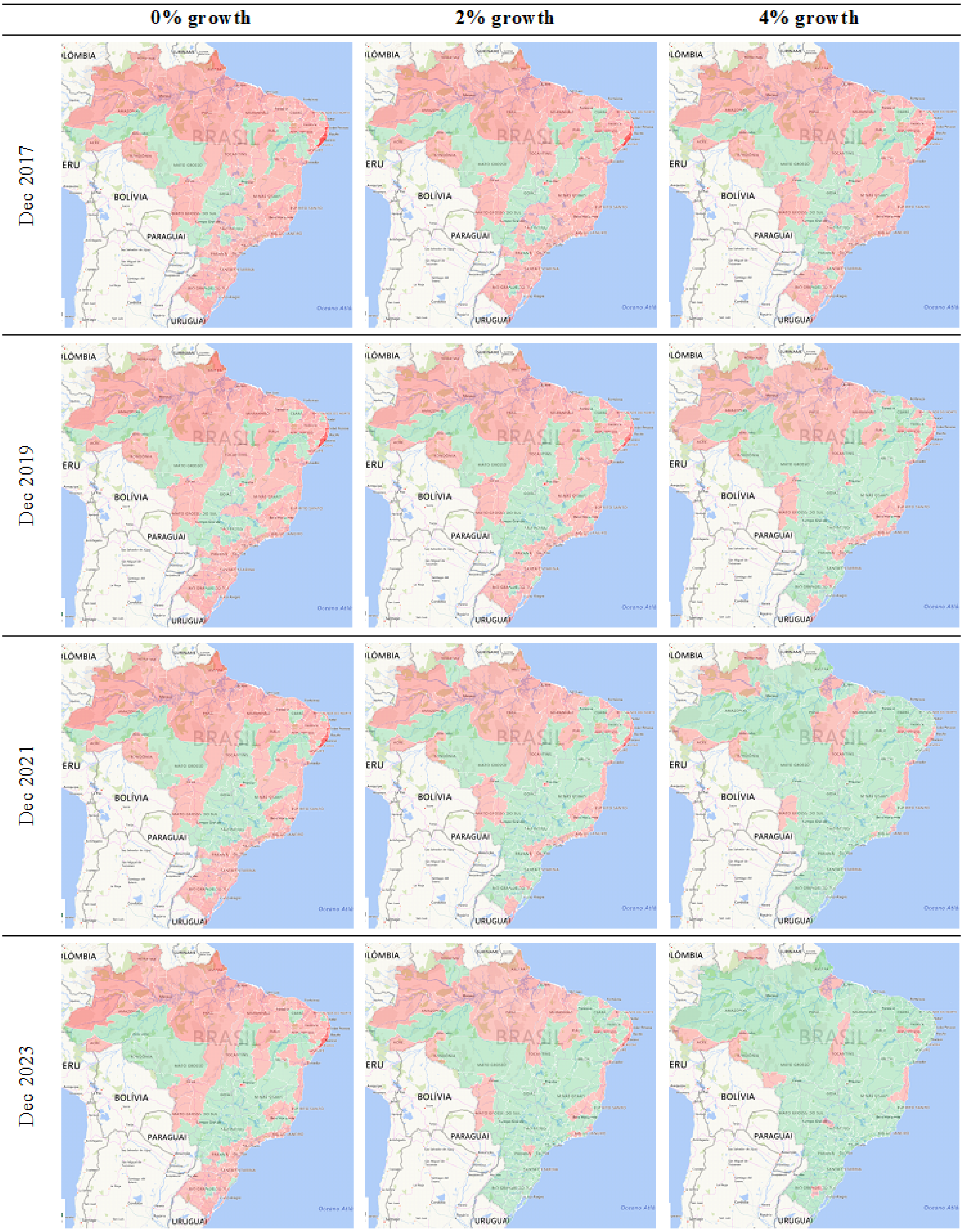
Figure 4: Cumulative employment rate changes for various industrial production growth scenarios on mesoregion level.



Source: elaborated by the author

Salvador: Salvador metropolitan area; Rio Jan: Rio de Janeiro metropolitan area; P.Alegre: Porto Alegre metropolitan area; B.Horizon.: Belo Horizonte metropolitan area SP: São Paulo metropolitan area

Figure 5: Accumulated employment rate variation for various industrial production growth scenarios on mesoregion level – Most populated mesoregions.



Source: elaborated by the author.
Original data from: CAGED

Figure 6: Dynamic comparison of the accumulated employment rate between different growth scenarios